Gauri Sharan

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Professional Experience

Software Dev Engineer (FTC), Whiteswan Identity Security – Remote (United States) Nov 2025 – March 2026

- Implemented Policy Enforcement Point (PEP), Policy Decision Point (PDP) and other Attribute-Based Access Control (ABAC) protocols in Agentic AI pipelines using JDBC-SpringBoot based web-apps and Python based MCP servers.

Trainee Intern - NBBL Technology, National Payments Corporation of India (NPCI) – Mumbai, India Mar 2025 – Aug 2025

- Built in-house LLM-based NL-to-SQL translator (Dolphin LLaMa quantized with LM Studio + Postgres) achieving substantial query accuracy; **won SolveX Hackathon** with **INR 200K prize**, demonstrating full-stack product development and team leadership.
- Developed time-series forecasting models (Prophet, ARIMA) for transaction trend prediction; automated reporting pipelines reducing manual effort, reports delivered to Google Pay, PhonePe, PayTM, and other major payment platforms.

AI Research Junior Assistant, IIT Roorkee – Delhi, India Dec 2024 – Feb 2025

- Researched RAG pipelines and Vision-Language Models for EdTech QA systems; evaluated LangChain, Pinecone, and LlamaIndex for production scalability.
- Collaborated with IIIT Delhi to develop RAG-based EdTech chatbot with curriculum-aligned content generation and retrieval mechanisms.

DevOps Intern, Whiteswan Identity Security – Remote (United States) Jul 2024 – Dec 2024

- Automated AWS Just-in-Time (JIT) provisioning using Boto3 and AWS Bash, improving security compliance and reducing manual provisioning time.
- Managed CI/CD deployments and optimized resource allocation across Docker, Kubernetes, EC2, IAM, reducing infrastructure costs while maintaining uptime.

Data Analyst Intern, NASSCOM – Noida, India Nov 2023 – May 2024

- Conducted statistical analysis on learner funnel data using Python (scikit-learn, matplotlib, seaborn, plotly); built dashboards tracking enrollment, completion, and certification conversion metrics.
- Delivered actionable performance insights to Digital India, NITI Aayog, and MeitY stakeholders, enabling data-driven improvements to digital literacy programs.

Projects

Insights Local AI github.com/gaurisharan/insights-local-ai

- SolveX Hackathon-winning analytics product: Integrated quantized Dolphin LLaMa models with Postgres backend (LM Studio) enabling NL-to-SQL querying for non-technical users. Achieved 95%+ query accuracy on financial transaction data.

Fine-tuned LLMs: SmolLM & Phi-2 huggingface.co/gauri-sharan/phi2-alpaca-lora-4bit, /SmolLM2-FT-MyDataset

- Fine-tuned SmolLM using SFT on SmolTalk dataset, improving response quality and coherence. Fine-tuned Microsoft Phi-2 with QLoRA (4-bit quantization) on Alpaca dataset, reducing VRAM requirements by 75% while maintaining performance.

C++ CUDA GPU Kernels for ML & HPC github.com/gaurisharan/cuda-ml-kernels

- Developed optimized CUDA kernels in C++ for parallelized matrix operations and training pipelines. Achieved 10x+ speedup on high-dimensional tensor computations vs CPU implementations.

RAG Document Chat Assistant github.com/gaurisharan/RAG-Advanced

- Built production-ready Retrieval-Augmented Generation chatbot using Pinecone vector store, LangChain orchestration, Mistral LLM, and HuggingFace embeddings for context-rich document Q&A with semantic search.

Real-Time Vision-NLP Multimodal System huggingface.co/gauri-sharan/test-two

- Developed bilingual (Hindi/English) image-to-text retrieval system integrating ByALDI and Qwen2VL for real-time keyword-based search and image classification in regional language applications.

Skills

Languages: Python, Java/SpringBoot, C, C++, SQL, JavaScript, Cedar
ML/AI: LLMs, RAG, Vision-Language Models, Fine-tuning (SFT/QLoRA), RNN/CNN/ANN, Time-Series Forecasting
Cloud & DevOps: AWS (EC2, IAM, JIT Provisioning), VertexAI, Azure ML, Docker, Kubernetes, Git
Tools & Frameworks: LangChain, Pinecone, LlamaIndex, Hugging Face, LM Studio, Boto3, FastAPI, MCP, A2A

Education

Dr. A.P.J. Abdul Kalam Technical University, AKTU – Masters in Computer Applications - 2027
Shobhit University – Bachelor of Science, Data Science - 2025 | 8.58 CGPA